

Limited Effects of Fertilization Practices on Microbial Community Structure and Greenhouse Gas-Related Functional Potential in Paddy and Field Soils

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Abstract: Fertilization practices are key management factors influencing greenhouse gas (GHG) emissions through soil microbial communities; however, the extent of these effects remains debated. We evaluated the impacts of different fertilization methods—standard, conventional, deep-placement, organic, and non-fertilization—on microbial community structure and GHG-related functional potential in paddy and field soils using 16S rRNA gene-based analysis and functional prediction. β -diversity analysis revealed that microbial communities were clearly separated by land-use type (paddy vs. field), with no significant differences among fertilization methods or soil depths within the same land-use type. α -diversity indices similarly showed no significant variation across fertilization treatments. LEfSe analysis identified selective shifts in specific taxa associated with GHG-related processes, including *Methanocella*, *Methylocystis*, *Nitrospira*, and *Rhodanobacteraceae*. Despite these taxonomic responses, PICRUSt2-based functional prediction revealed highly consistent metabolic module distributions across fertilization treatments, corroborated by stable patterns of functional genes involved in methanogenesis, methane oxidation, nitrification, and denitrification. These results demonstrate a decoupling between taxonomic shifts and functional potential, suggesting that fertilization practices exert limited influence on GHG-related functional capacity. Instead, functional redundancy within microbial communities, together with land-use type and environmental constraints, appears to maintain functional stability.

Keywords: Fertilization management, Functional redundancy, Greenhouse gas, Soil microbial community, Soil microbial functional potential

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Introduction

Agricultural lands are major anthropogenic sources of greenhouse gases (GHGs), including methane (CH₄) and nitrous oxide (N₂O), and exert a significant influence on global climate change. The production and consumption of these GHGs are primarily regulated by microbe-mediated biogeochemical processes such as methanogenesis, methane oxidation, nitrification, and denitrification. The structural and

functional characteristics of soil microbial communities act as key regulators of agricultural GHG emissions. Therefore, elucidating the impacts of agricultural management practices on soil microbial communities and their functional potential is essential for establishing sustainable agricultural strategies to mitigate climate change. Among agricultural land-use types, paddy and field soils provide distinctly contrasting ecological environments. Paddy soils are characterized by limited oxygen availability and dominant reductive conditions due to flooding management, whereas field soils are relatively aerobic and frequently exposed to mechanical disturbances like tillage. These hydrological and physicochemical differences significantly influence the composition and assembly processes of microbial communities. Numerous previous studies have reported that flooding status is the dominant factor driving microbial community differentiation, surpassing the influence of crop species [6].

Soil microbial communities can also differentiate along environmental gradients within the vertical soil profile. Generally, variations in oxygen, organic carbon, and nutrients across depths induce microbial niche differentiation. However, in agricultural lands, solute movement via irrigation, vertical infiltration, and physical mixing by tillage can mitigate this vertical heterogeneity. Indeed, an increasing number of reports suggest that differences in microbial communities according to depth are limited in intensively managed paddy and field soils [12]. Fertilization practices are also critical management factors that can influence soil microbial communities. Fertilization alters nutrient availability and stoichiometric constraints in the soil, potentially inducing increases or decreases in specific microbial taxa or functional groups. However, findings regarding the impact of fertilization on overall microbial diversity or community structure remain inconsistent across studies. Even in long-term fertilization experiments, its effects are often reported to be limited compared to overarching environmental factors such as soil moisture status or land-use type [4]. This suggests that the effects of fertilization may manifest selectively at the level of specific taxa or functional niches rather than the community as a whole. Recently, there has been increasing interest in the "decoupling" between shifts in microbial community composition and functional potential. However, it remains insufficiently understood how this phenomenon manifests across different agricultural land-use types. In particular, whether taxonomic shifts observed due to fertilization extend to actual changes in the GHG-related functional gene pool remains a significant research gap. Accordingly, this study comprehensively evaluated the effects of land-use type, soil depth, and fertilization methods on soil microbial community structure and GHG-related functional potential in paddy and field soils. Through 16S rRNA gene-based community analysis and Phylogenetic Investigation of Communities by Reconstruction of Unobserved States 2 (PICRUSt2)-based functional prediction, we aimed to verify (i) the patterns of microbial community differentiation between paddy and field soils, (ii) community stability within the vertical soil profile, (iii) changes in microbial diversity and community structure according to fertilization methods, and (iv) whether taxonomic-level changes translate into shifts in functional potential at the GHG metabolic pathway level.

Results and Discussion

Community Differentiation by Land-Use Type and Stability within Vertical Profiles

In this study, the microbial community structures of paddy and field soils were compared using β -diversity analysis, which revealed statistically significant differences between the two land-use types (Fig. 1). This indicates that the distinct soil environmental conditions shaped by land use are major determinants of microbial community composition. Previous research has reported that paddy soils, influenced by flooding management, experience more restricted oxygen availability compared to field soils, leading to distinct redox gradients and unique chemical properties (e.g., pH fluctuations) that create microbial habitats differentiated from those of aerobic upland fields [6,11]. Interestingly, within each land-use type, the analysis of community structure across soil depths showed no significant vertical differentiation in either paddy or field soils ($p>0.05$; Supplementary Fig. S1 and Table S1). This absence of vertical stratification suggests that the management practices employed in each agricultural system induce microbial homogenization throughout the soil profile. In paddy soils, the limited effect of depth can be explained by hydrological connectivity under irrigation management. Vertical percolation and preferential flow pathways facilitate the active downward transport of dis-

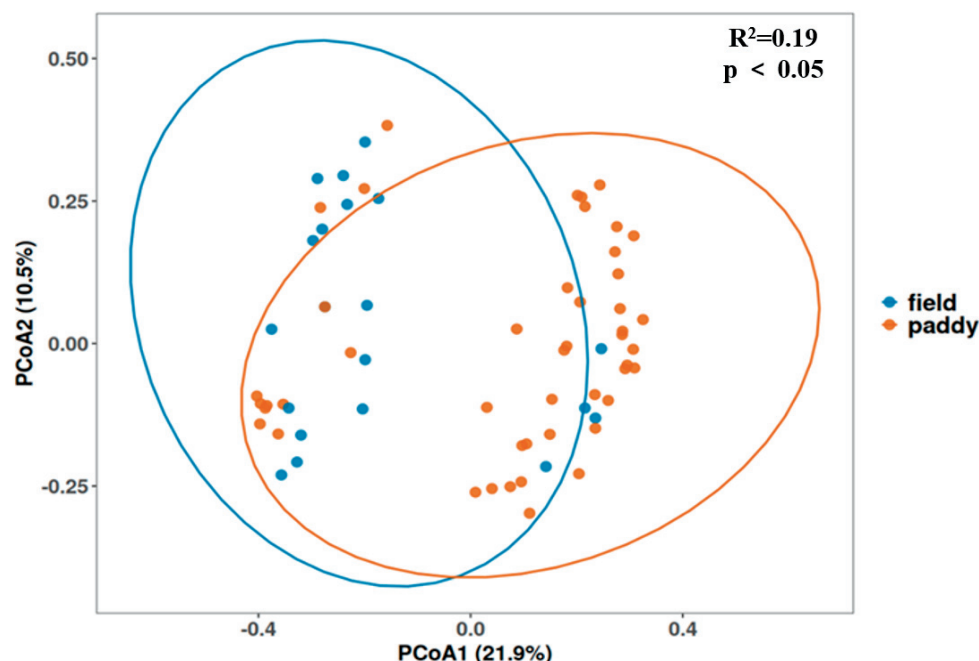


Fig. 1. Principal coordinate analysis (PCoA) of soil microbial community structure in paddy and field soils based on Bray–Curtis dissimilarity.

solved organic matter (DOC) and soluble nutrients from the surface to deeper layers [22, 9]. This mitigation of resource imbalances across depths is thought to inhibit the vertical segregation of microbial communities [12]. Similarly, the microbial community structure in field soils did not differ significantly across depth intervals ($p > 0.05$), likely due to mechanical disturbances such as tillage and plowing, which are common in non-flooded croplands. Periodic soil inversion and mixing disrupt the development of vertical environmental gradients and physically blend the microbial populations, maintaining community composition in the subsoil similar to that of the surface layer [2]. Taken together, these findings suggest that irrigation management in paddies and mechanical tillage in fields serve as major drivers of microbial community stability across soil profiles through hydrological and physical pathways, respectively.

Differences in Microbial Diversity by Fertilization Practices

Analysis of the impact of fertilization practices on microbial diversity in both paddy and field soils revealed that α -diversity (Shannon and Chao1 indices) did not exhibit statistically significant differences among treatments (Fig. 2). In both paddy and field soils, the Kruskal–Wallis test confirmed no significant influence of fertilization practices on any diversity indices (Table 1), suggesting that the structural diversity within the microbial communities remained robustly stable under the various fertilization regimes tested in this study. β -diversity analysis (principal coordinates analysis (PCoA) based on Bray–Curtis distance) further supported this stability, as no distinct separation of community structures was observed according to fertilization in either paddy or field soils (Fig. 3). This lack of systemic shifts was statistically reinforced by ANOSIM results (Paddy: $R=0.05$, $p > 0.05$; Field: $R=0.00$, $p > 0.05$) and pairwise Permutational Multivariate Analysis of Variance (PERMANOVA), which consistently yielded no significant differences among treatments ($p > 0.05$; Supplementary Table S2). These findings align with previous research indicating that long-term agricultural management can have limited effects on the overall structural assembly of soil bacterial communities [4]. In summary, the impact of fertilization practices was not manifested as systemic changes in community-wide diversity or structure. Consequently, Section 3 utilizes Linear Discriminant Analysis Effect Size (LEfSe) analysis to identify selective responses at the specific taxonomic level, which may provide deeper insights into the subtle modulations within the functional microbial niches.

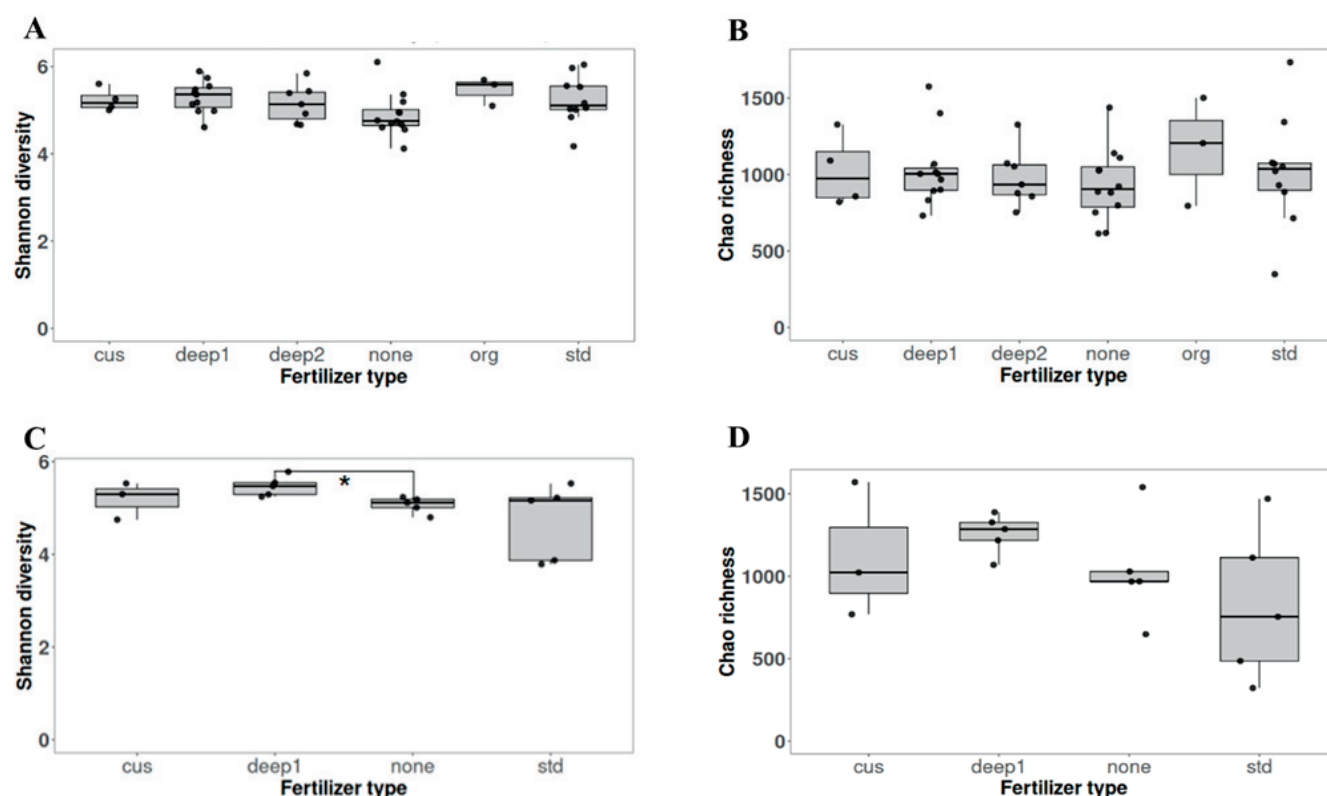


Fig. 2. Microbial alpha diversity indices across different fertilization practices in paddy and field soils: (A) Shannon diversity in paddy soil, (B) Chao1 richness in paddy soil, (C) Shannon diversity in field soil, (D) Chao1 richness in field soil.

Differentially Abundant Taxa Identified via LEfSe Analysis relative to Standard Fertilization

Analysis of the impact of fertilization practices using LEfSe relative to standard fertilization (std) revealed significant differences in the relative abundance of specific microbial taxa depending on the treatment (Fig. 4). In paddy soils (Fig. 4A), among the differentially detected taxa, *Methanocella*, which is associated with methanogenesis, significantly increased in the unfertilized treatment (none) [14]. Conversely, the methanotroph *Methylocystis* showed a decreasing trend in the reduced deep fertilization treatment (deep2) [1] (Fig. 4). In field soils (Fig. 4B), *Nitrospira*, a key taxon involved in nitrification, was significantly enriched in the deep fertilization treatment (deep1) compared to the standard practice [7]. In contrast, *Rhodanobacteraceae*, which is associated with the nitrogen cycle, significantly decreased in the same treatment (deep1) [10] (Fig. 4). Interestingly, the aforementioned taxa that exhibited fluctuations in response to fertilization practices are known to play pivotal roles in soil methane and nitrogen cycling—specifically, in GHG metabolism. These shifts at the individual taxonomic level suggest that fertilization practices can exert selective influences on GHG-related microbial niches. To investigate how these microscopic changes relate to the overall functional potential of the soil, the next section compares the distribution of functional genes associated with GHG metabolic pathways to evaluate consistency with the observed taxonomic shifts.

Table 1. Kruskal–Wallis test results for alpha diversity indices (Chao1 and Shannon) across fertilization practices in field and paddy soils

Landuse	Metric	χ^2	df	p value
Field	Chao1	1.754167	3	0.624959
Field	Shannon	4.741667	3	0.191721
Paddy	Chao1	3.312751	5	0.65189
Paddy	Shannon	7.863416	5	0.163925

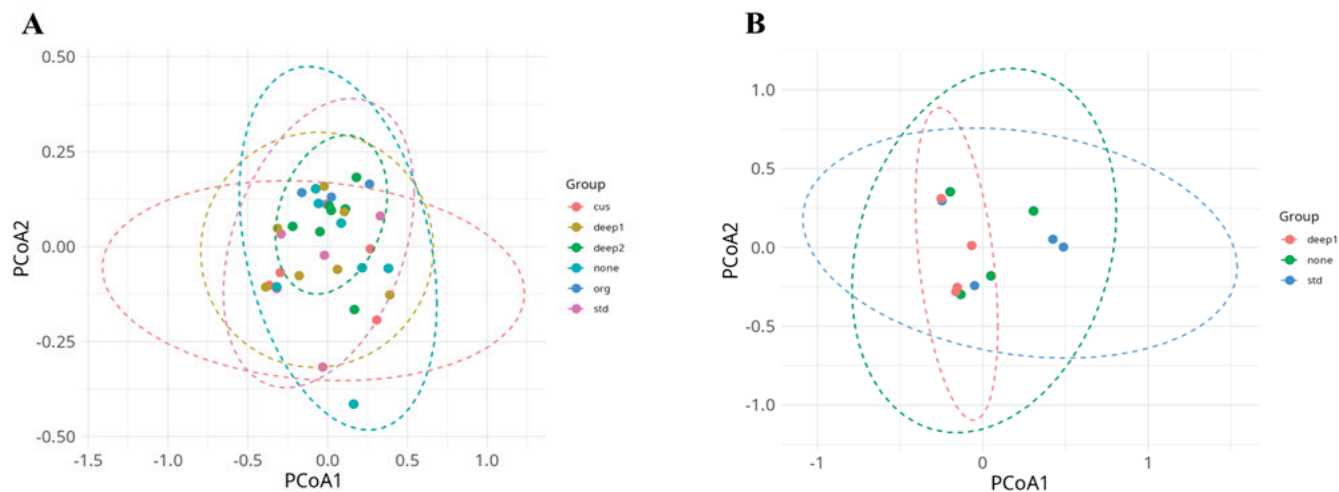


Fig. 3. Principal coordinate analysis (PCoA) of soil microbial community structure based on Bray–Curtis dissimilarity across different fertilization practices: (A) paddy soil, (B) field soil.

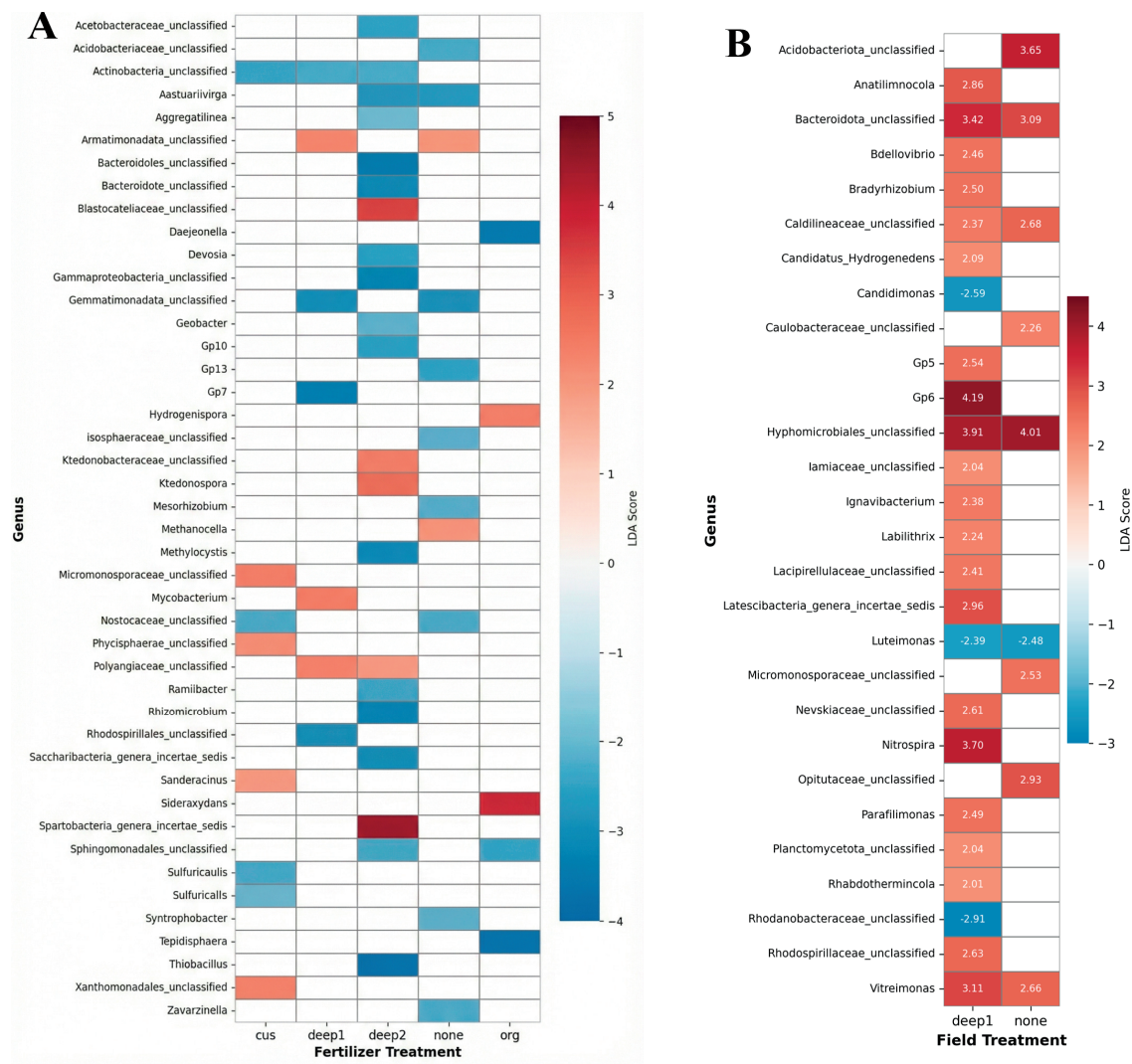


Fig. 4. LEfSe analysis of differentially abundant microbial genera across various fertilization practices: (A) paddy soil, (B) field soil. Genus-level groups designated as 'Gp' (e.g., Gp5, Gp6) represent members of the phylum Acidobacteria. Only taxa with an LDA score > 2.0 are shown.

Robustness of GHG-related Functional Potential to Fertilization Practices

To evaluate the impact of fertilization practices on the functional characteristics of the soil microbiome, we first quantified the total potential—a representative index for the cumulative capacity of metabolic pathways—for four core pathways involved in CH_4 and N_2O metabolism: nitrification, denitrification, methanogenesis, and methane oxidation. In both paddy and field soils, no statistically significant differences in total potential were detected among fertilization practices (Kruskal–Wallis test, $p > 0.05$; Fig. 5, Supplementary Table S3). Furthermore, pairwise comparisons based on Dunn's post-hoc test with Benjamini–Hochberg correction showed that all treatments were grouped under the same letter ('a'), statistically confirming that fertilization practices did not significantly alter the aggregate indicators at the pathway level (Fig. 5, Supplementary Table S4). Subsequently, to determine whether this macroscopic stability masked any underlying shifts at the gene level, the relative abundances of GHG-related functional genes predicted via PICRUSt2 were log-scaled and compared (Supplementary Fig. S2). Consistent with the module-level observations, the overall distribution patterns of individual functional genes remained highly similar across fertilization treatments in both paddy and field soils, despite the differential abundance of specific taxa observed in the preceding LEfSe analysis (Supplementary Fig. S2). It should be noted that PICRUSt2 provides predictions of functional gene potential rather than direct measurements of in situ metabolic activity or GHG flux. Therefore, empirical validation through gas flux measurements or metatranscriptomic approaches would further strengthen the conclusions drawn here. These findings suggest that land-use type is a primary driver of soil microbial community structure and associated GHG-related functional potential, with fertilization practices exerting comparatively limited and taxon-selective effects. From a practical standpoint, this implies that broad-scale GHG mitigation strategies in agricultural soils may be more effectively achieved through land-use management decisions—such as conversion between paddy

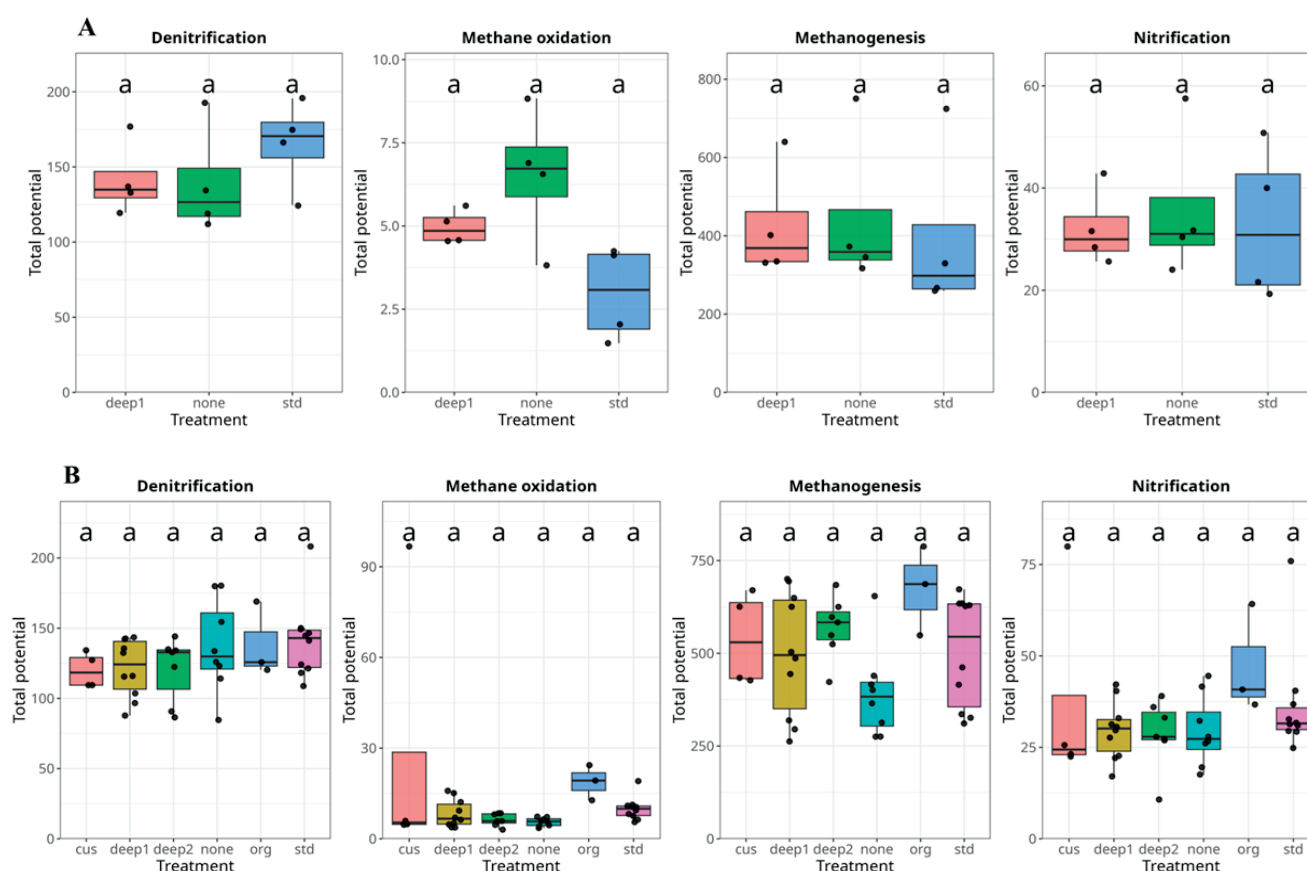


Fig. 5. Comparison of the total functional potential for key greenhouse gas (GHG) metabolic pathways across fertilization practices in field soil (A) and paddy soil (B).

and upland systems—rather than through optimization of fertilization regimes alone. Nevertheless, the taxon-level shifts observed in response to fertilization (e.g., *Methanocella*, *Nitrospira*) indicate that targeted management of specific niches may still offer opportunities for fine-scale modulation of GHG-cycling microorganisms. In summary, while fertilization practices can induce selective shifts at specific taxonomic levels, these changes do not consistently extend to shifts in the community-wide gene pool or module-level functional indicators. This taxonomic-functional decoupling [3], where taxonomic variation does not translate into functional differentiation, can be interpreted alongside the high level of functional redundancy within the soil microbial community, wherein multiple phylogenetically distinct taxa share overlapping metabolic capabilities, allowing other community members to compensate when specific taxa such as *Methanocella* or *Nitrospira* fluctuate in relative abundance [4]. Moreover, it suggests that higher-level environmental constraints—such as flooding conditions in paddies or the stable physico-chemical environment of field soils—may exert a stronger influence on the maintenance of microbial functional configurations than fertilization management [8]. However, since these results are based on 16S rRNA-based predictions (PICRUSt2), future studies incorporating shotgun metagenomic sequencing and transcriptomics, alongside actual environmental data (e.g., pH, C/O ratios, Organic matter), are necessary to precisely validate the correlation between predicted functional potentials and empirical GHG emission rates.

Materials and Methods

Soil sample collection

Soil samples were collected from May to August at an experimental field located in Deokjin-gu, Jeonju-si, Jeonbuk, Korea (approx. 35.835° N, 127.050° E). The experimental design consisted of six fertilization treatments: standard fertilization (std), no fertilization (none), deep fertilization (deep1), deep fertilization with reduced rates (deep2), customary fertilization (cus), and organic fertilization (org). At each treatment plot, soil samples were collected at three distinct depths: topsoil (top, 0–20 cm), middle soil (middle, 20–40 cm), and subsoil (sub, 40–60 cm). Three independent experimental plots (n=3) were sampled per treatment to ensure biological replicates. The collected samples, intended specifically for DNA extraction, were cleared of plant residues and stones and immediately transported to the laboratory in an ice box (maintained below 4°C). Subsequently, the soil was aliquoted into sterile tubes and stored at –80°C until further processing for DNA extraction.

Total DNA extraction and 16S rRNA gene amplicon sequencing

Total DNA was extracted from the soil samples using the ExgeneTM Soil DNA Mini Kit (GeneAll Biotechnology, Seoul, Korea). The concentration of the extracted DNA was quantified using a Qubit 2.0 fluorometer (Thermo Fisher Scientific, Waltham, MA, USA) with the Qubit dsDNA HS Assay Kit (Thermo Fisher Scientific, MA, USA). The purified DNA was stored at –80°C until subsequent analysis. The 16S rRNA gene libraries were prepared according to Illumina's 16S Metagenomic Sequencing Library Preparation Protocol. The V3–V4 hypervariable regions of the 16S rRNA gene were amplified using the primers 341F (5'-CCTACG GNGGCWGCAG-3') and 805R (5'-GACTACHVGGGTATCTAATCC-3') with TaKaRa EX Taq Hot Start Version (Takara Bio, Shiga, Japan). The resulting PCR products were purified using HiAccuBeads (AccuGene, Korea), followed by the attachment of Illumina index adapters (barcodes). The indexed amplicons underwent an additional purification step before final pooling. Sequencing was conducted on the Illumina MiSeq platform (2×300 bp paired-end reads) at Macrogen Inc. (Seoul, Korea).

Data processing

To ensure an equal sequencing effort across all samples, the raw sequence dataset was initially subsampled to 10,000 reads per sample. Subsequently, the reads were processed using Mothur software (v1.48.2) [5]. After merging paired-end reads, low-quality reads were removed. The resulting reads were aligned to the SILVA reference database (v138.2) [16], and chimeric sequences were

identified and removed with VSEARCH (v2.17.1) [17]. Operational taxonomic units (OTUs) were clustered at 97% similarity using the OptiClust algorithm [14] and classified against the Ribosomal Database Project (RDP) 16S rRNA training set (v19) [5].

Functional prediction

Functional profiles were inferred from 16S rRNA gene sequences using PICRUSt2 (v2.6.2) [2]. Representative sequences and the sample-by-OTU abundance table generated in mothur were used as inputs. The PICRUSt2 workflow (picrust2_pipeline.py) was applied to place sequences into a reference phylogeny, predict gene family abundances at the level of KEGG Orthologs (KOs), and summarize predicted functions at the pathway/module level. Stratified outputs were additionally generated to decompose predicted functions into OTU-level contributions. For GHG-related functional analyses, KO sets associated with methanogenesis, methane oxidation, nitrification, and denitrification were curated based on KEGG module definitions (accessed on December 2025).

Statistical analysis

Normality was evaluated using the Shapiro–Wilk test in R (v4.5.2) for each comparison, and non-parametric tests were applied when normality assumptions were not met. α -diversity indices (Shannon and Chao1) were calculated in Mothur (v1.48.2). Differences in α -diversity among treatments were assessed using the Kruskal–Wallis test, and p -values were adjusted for multiple testing using the Benjamini–Hochberg false discovery rate (FDR) procedure using the FSA package (v0.10.0) in R [14]. Because depth-specific biological replication ($n=3$) was incomplete for some fertilization treatments, α -diversity comparisons for the fertilization $\times$ depth interaction were not performed; the final sample sizes included in each group are summarized in Table S1. β -diversity was computed using Bray–Curtis dissimilarity and visualized by PCoA, then Group differences in community composition were tested by Analysis of Similarities (ANOSIM) and PERMANOVA (perm. 999) in R using the vegan package (v2.7-2) [15]. Pairwise PERMANOVA was additionally conducted, and p -values were adjusted using the Benjamini–Hochberg FDR method. Taxonomic differential abundance was assessed using the LEfSe() function implemented in Mothur (v1.48.2). The Mothur-generated OTU abundance table and the corresponding design file containing group assignments were used as inputs, and LEfSe was executed using the default settings. Comparisons of PICRUSt2-predicted functional profiles (KO- and module/pathway-level) among fertilization treatments were performed using predicted relative abundances. For visualization, KO-level profiles were transformed using $\log(x+1)$ and displayed as a heatmap with hierarchical clustering. Module/pathway-level total potential was computed separately for paddy and field soils as a weighted index derived from PICRUSt2 outputs by multiplying taxon relative abundance by genome function counts. GHG-related KOs/modules were compared across land-use and fertilization treatments using the Kruskal–Wallis test. Post-hoc comparisons were conducted using Dunn’s test, and group labeling was summarized using compact letter display (CLD) using rcompanion package (v2.5.1) [12] in R. Statistical significance was set at $p<0.05$.

Data Availability: All data are available in the main text or in the Supplementary Information.

Author Contributions: YJ collected the soil sample, NJ collected the data, performed the bioinformatic analysis, conducted statistical analysis and wrote the first manuscript, JH and JO reviewed the manuscript, TU and DL designed the research and finalized the manuscript.

Notes: The authors declare no conflict of interest

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Additional Information:

Supplementary information The online version contains supplementary material available at <https://doi.org/10.5338/KJEA.2026.45.08>

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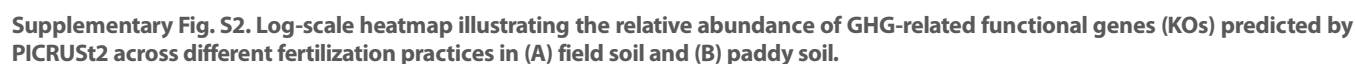


Table S1. Pairwise PERMANOVA results comparing microbial community structures between soil depth intervals in paddy and field soils

Land-use	Group 1	Group 2	R ²	F	p	p_BH
Paddy	Top	Sub	0.03	0.9	0.45	0.45
Paddy	Top	Middle	0.07	1.5	0.1	0.32
Paddy	Middle	Sub	0.05	1.23	0.22	0.33
Field	Top	Sub	0.15	1.78	0.1	-

Table S2. Pairwise PERMANOVA results for soil microbial community structures among different fertilization practices in paddy and field soils

Land-use	Group 1	Group 2	R ²	F	p	p_BH
Paddy	std	deep2	1.13	0.09	0.3	0.98
Paddy	std	cus	0.45	0.05	0.96	0.98
Paddy	std	deep1	0.36	0.03	0.98	0.98
Paddy	std	none	0.6	0.05	0.84	0.98
Paddy	std	org	0.94	0.11	0.47	0.98
Paddy	deep2	cus	1.13	0.11	0.35	0.98
Paddy	deep2	deep1	0.98	0.08	0.39	0.98
Paddy	deep2	none	0.86	0.07	0.58	0.98
Paddy	deep2	org	1.14	0.13	0.28	0.98
Paddy	cus	deep1	0.44	0.05	0.88	0.98
Paddy	cus	none	0.45	0.05	0.92	0.98
Paddy	cus	org	0.86	0.15	0.46	0.98
Paddy	deep1	none	0.58	0.05	0.81	0.98
Paddy	deep1	org	0.75	0.09	0.58	0.98
Paddy	none	org	0.72	0.08	0.75	0.98
Field	none	std	0.89	0.13	0.5	0.59
Field	none	deep1	0.84	0.12	0.59	0.59
Field	std	deep1	1.45	0.19	0.27	0.59

Table S3. Kruskal–Wallis tests comparing the total functional potential of GHG-related metabolic modules among fertilization practices in field and paddy soils.

Land-use	Pathway module	p-value	Land-use	Pathway module	p-value
Field	denitrification	0.437393043	Paddy	denitrification	0.525278401
	methane_oxidation	0.049787068		methane_oxidation	0.017393832
	methanogenesis	0.397294713		methanogenesis	0.112613751
	nitrification	0.841072914		nitrification	0.257566914

Table S4. Pairwise comparisons of the total functional potential for GHG-related metabolic modules among fertilization practices in field and paddy soils based on Dunn's post-hoc test.

Land-use	Module	Group 1	Group 2	Z statistic	p-value (unadj)	p-value (adj)
Field	Methane oxidation	none	std	2.353	0.0186	0.0558
Paddy	Methane oxidation	none	std	-2.535	0.0113	0.0563
Paddy	Methane oxidation	deep2	org	-2.616	0.0089	0.0668
Paddy	Methane oxidation	none	org	-3.1	0.0019	0.069
Paddy	Methane oxidation	deep1	org	-2.291	0.022	0.0824
Paddy	Methanogenesis	none	org	-2.714	0.0066	0.0997
Paddy	Methane oxidation	cus	org	-2.108	0.035	0.1051

Land-use	Module	Group 1	Group 2	Z statistic	p-value (unadj)	p-value (adj)
Field	Methane oxidation	deep1	std	1.765	0.0776	0.1163
Paddy	Methane oxidation	deep2	std	-1.843	0.0653	0.1633
Paddy	Nitrification	deep2	org	-1.918	0.0551	0.2066
Paddy	Nitrification	deep1	org	-2.027	0.0427	0.2135
Paddy	Methanogenesis	cus	none	1.581	0.1139	0.2442
Paddy	Methanogenesis	deep1	none	1.594	0.111	0.2774
Paddy	Methanogenesis	deep2	none	2.05	0.0403	0.3025
Paddy	Nitrification	cus	org	-2.046	0.0408	0.306
Paddy	Methanogenesis	org	std	1.618	0.1057	0.317
Paddy	Methane oxidation	org	std	1.362	0.1732	0.3247
Paddy	Methane oxidation	cus	std	-1.206	0.228	0.342
Paddy	Methane oxidation	deep1	none	1.246	0.2128	0.3547
Field	Methanogenesis	deep1	none	1.177	0.2393	0.359
Paddy	Methane oxidation	deep1	std	-1.367	0.1716	0.3677
Paddy	Methanogenesis	none	std	-1.628	0.1035	0.388
Paddy	Methane oxidation	deep2	org	-1.125	0.2606	0.4343
Paddy	Methane oxidation	cus	org	-1.138	0.2549	0.478
Field	Methane oxidation	deep1	org	-1.643	0.1004	0.5021
Paddy	Methanogenesis	deep1	org	-1.643	0.1004	0.5021
Paddy	Nitrification	deep1	std	-1.075	0.2822	0.5291
Paddy	Nitrification	none	std	-1.13	0.2585	0.554
Field	Methane oxidation	deep1	none	-0.588	0.5563	0.5563
Paddy	Nitrification	deep2	std	-0.955	0.3398	0.5663
Paddy	Nitrification	cus	std	-1.199	0.2306	0.5766
Paddy	Methane oxidation	cus	none	0.799	0.4245	0.5788
Paddy	Nitrification	org	std	1.296	0.1949	0.5848
Paddy	Nitrification	none	org	-2.052	0.0402	0.6027
Field	Denitrification	none	std	-1.275	0.2024	0.6072
Field	Denitrification	deep1	none	0.49	0.6239	0.6239
Field	Denitrification	deep1	std	-0.784	0.4328	0.6492
Paddy	Methane oxidation	deep2	none	0.568	0.5699	0.6576
Paddy	Methane oxidation	deep1	deep2	0.603	0.5468	0.6835
Field	Methanogenesis	none	std	1.177	0.2393	0.7179
Paddy	Methanogenesis	deep2	std	0.586	0.5579	0.7607
Field	Nitrification	deep1	none	-0.294	0.7686	0.7686
Paddy	Methanogenesis	deep1	deep2	-0.619	0.5359	0.8038
Paddy	Methane oxidation	cus	deep2	0.311	0.7557	0.8096
Paddy	Denitrification	cus	std	0.331	0.7409	0.8549
Paddy	Denitrification	deep1	std	-0.743	0.4575	0.8578
Paddy	Methane oxidation	cus	deep1	-0.172	0.8633	0.8633
Paddy	Methanogenesis	cus	deep1	-0.172	0.8633	0.8633
Paddy	Denitrification	deep2	none	-0.835	0.4035	0.8647
Paddy	Denitrification	deep2	org	-0.641	0.5214	0.8689
Paddy	Methane oxidation	cus	deep1	0.358	0.7202	0.9002
Paddy	Denitrification	cus	deep1	-0.276	0.7829	0.9033
Paddy	Denitrification	deep2	std	-1.328	0.1842	0.9209
Paddy	Nitrification	org	std	-0.322	0.7475	0.9344

Land-use	Module	Group 1	Group 2	Z statistic	p-value (unadj)	p-value (adj)
Paddy	Denitrification	deep1	none	-1.01	0.3127	0.9381
Paddy	Methanogenesis	cus	deep2	-0.149	0.8818	0.9448
Paddy	Nitrification	cus	deep2	-0.381	0.7033	0.9591
Paddy	Denitrification	none	std	-0.468	0.6396	0.9594
Paddy	Methanogenesis	deep1	std	-0.036	0.9709	0.9709
Paddy	Nitrification	cus	none	-0.283	0.7773	0.9716
Paddy	Nitrification	deep1	none	0.116	0.9077	0.9725
Paddy	Denitrification	cus	org	-0.854	0.3932	0.983
Paddy	Nitrification	deep1	deep2	-0.021	0.983	0.983
Field	Methane oxidation	none	org	-0.015	0.988	0.988
Paddy	Denitrification	none	org	-0.015	0.988	0.988
Paddy	Denitrification	deep1	deep2	-0.095	0.9247	0.9907
Field	Methane oxidation	deep1	none	0	1	1
Field	Nitrification	none	std	0.588	0.5563	1
Field	Nitrification	deep1	std	0.294	0.7686	1
Paddy	Denitrification	deep1	std	-1.568	0.117	1
Paddy	Denitrification	cus	std	-1.461	0.1442	1
Paddy	Denitrification	cus	none	-1.048	0.2945	1
Paddy	Denitrification	cus	deep2	-0.334	0.7381	1
Paddy	Nitrification	cus	deep1	-0.386	0.6996	1
Paddy	Nitrification	deep2	none	0.127	0.8993	1